EDC · PAPER

# EDC practice test: marking guide

Practice questions · Day 08 · Sat 31 Oct · pack EPT

**How to use this:** Sit the paper first ("Resources/Practice test_ENGE2003.pdf"): 90 minutes, timed, open book, calculator, no peeking at this guide. Then put your pen down and mark it question by question, opening each `<details>` block only when you are ready to compare. Be strict: award a mark only if your script shows the line that earns it. Write your score per question in the table under "After marking" and use that table to plan your revision.

The paper has 8 questions and 50 marks. The formula table is on page 2 of the paper (before Q1), not at the back. Every circuit value in this guide was read from the paper's own figures and matches the values stated in your Test 1 Reference (p. 17–21). Every number was recomputed in Python.

## Before you start

**Time budget.** 90 min for 50 marks is **1.8 min per mark**. Leave about 5 min at the end to check.

| Question | Marks | Target time | Running clock |
|---|---|---|---|
| Q1 MCQ | 5 | 5 min | 0:05 |
| Q2 exponential model | 5 | 7 min | 0:12 |
| Q3 three diodes, PWL | 8 | 14 min | 0:26 |
| Q4 Zener design | 5 | 8 min | 0:34 |
| Q5 clipper | 5 | 8 min | 0:42 |
| Q6 load line | 6 | 12 min | 0:54 |
| Q7 Thevenin + BJT | 10 | 16 min | 1:10 |
| Q8 two BJTs | 6 | 12 min | 1:22 |
| Check | | 8 min | 1:30 |

Q1 is quick. Bank the time you save there for Q6, because it needs a ruler and careful graph reading.

**What earns method marks on this paper.**

- Write the formula in symbols first, then substitute, then the answer with units. For example, $\Delta V_D = nV_T \ln(I_2/I_1)$, then numbers, then "= 0.179 V". If the number is wrong but the formula and substitution are right, you still get most of the marks.

- State the model you are using in words: “third PWL model: each diode = ideal diode + 0.6 V + 0.2 kΩ”, “active mode: $V_{BE} = 0.7$ V, $I_C = \beta I_B$”, “ideal diode”.
- “Redraw” parts (Q3(a), Q7(b)) are worth 2 and 4 marks. Draw them neatly and label every element value and every node you later use ($V_A$, $V_B$, $I_B$, $\beta I_B$).
- “Verify” parts (Q3(b)) need a stated conclusion: “I = 15.3 µA > 0, so all three diodes conduct.” A number without the sentence loses the mark.
- When the question tells you to assume a state (Q3, Q7, Q8), check it at the end in one line: current > 0, $V_{CE} > V_{CE,sat}$.

**Before moving on from each question, check these:**

1. Units. Are kΩ and mA paired? Is µA written as µA, not mA?
2. Sign and direction. Do the current arrows on your sketch match the signs you used?
3. Is the state-assumption check written down?
4. Is the final answer underlined or boxed, with units?